

THE PROPERTIES AND APPLICATIONS OF A FINITE CONFIDENCE SEQUENCE FOR THE MEAN OF A NORMAL POPULATION WITH UNKNOWN VARIANCE

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SUMMARY

The properties of a confidence sequence for the unknown mean μ of a normal distribution with the variance also unknown are considered. It is shown how this sequence can be used to test sequentially the hypothesis $\mu = \mu_0$ against a two-sided alternative. Monte Carlo experiments indicate that this test can be considered to be a strong competitor against existing sequential t-tests and in some cases gives better results. It is further shown how a confidence interval for μ can be computed after termination of the sequential test. Special attention is paid in this paper to application of the proposed confidence sequence in practice.

1. Introduction

Let $X_1, X_2, \dots$ be independent observations from a normal population with mean μ and variance σ^2 . Consider the problem of constructing a sequence $\{I_n\}$ of intervals such that for all μ

$$(1) \quad P_{\mu}(\mu \in I_n \text{ for all } m \leq n \leq \ell) \geq 1 - \alpha$$

m and ℓ being positive integers. The sequence $\{I_n\}$ is called a $(1 - \alpha)$ -confidence sequence for μ . Applications of such a sequence are found in sequential interval estimation as well as sequential testing concerning μ . In Sections 5 and 6 applications will be considered.

In the case where σ is known, confidence sequences for μ were constructed by Paulson (1969) and Armitage (1957), both for finite ℓ . The case where σ is unknown, was considered by Robbins (1970), who proposed a confidence sequence for μ , but for $\ell = \infty$. The case $\ell < \infty$ is of more practical importance. Here a confidence sequence based on the repeated significant t-test (RST-plans) as described by Armitage (1975) can be used. This $(1 - \alpha)$ -confidence sequence has bounds

$$(2) \quad \bar{X}_n \pm s_n t_{n-1}(\alpha') / \sqrt{n}, \quad n = 2, \dots, \ell,$$

where $\bar{X}_n = \sum_{i=1}^n X_i / n$, $s_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)$ and $t_{n-1}(\alpha')$ is the critical value at

significant level $\alpha' < \alpha$ of the Student t two-sided test (α' depends on ℓ and can be obtained from Table 6.4 in Armitage (1975) for selected values of ℓ). Siegmund (1977) also obtained asymptotic approximations to α' . The disadvantage of this sequence is that tables containing values of $t_{n-1}(\alpha')$, with α' usually very small, are not widely available.

A more practical sequence follows from a suggestion made by Samuel-Cahn (1974b). This sequence has bounds

$$(3) \quad \bar{X}_n \pm s_n z_\alpha \ell^2/n, \quad n = 2, \dots, \ell,$$

where z_α is given by $\Phi(z_\alpha) = 1 - \frac{1}{4}\alpha$, Φ the standard normal distribution function.

This interval is the same as suggested by Paulson, but replacing σ by its sample estimate s_n . Intuitively one would feel that the coverage probability of the interval with bounds (3), can be $1-\alpha$ when ℓ is large. This fact was indeed proven by Swanepoel (1977) (theorem 2.2) for a more general case.

In this paper we want to investigate the confidence sequence (with bounds given by (3) and ℓ finite) with respect to the following:

- (a) A method of adapting the sequence in order to obtain a coverage probability $1-\alpha$.
- (b) A method for using it in testing the hypothesis $H_0: \mu = \mu_0$, sequentially.
- (c) The properties of this sequential test and comparisons with the fixed sample size t -test as well as other sequential t -tests.

2. The coverage probability of the confidence sequence for μ

We have for I_n with the bounds given in (3), that (Swanepoel, 1977)

$$\lim_{\ell \rightarrow \infty} P_\mu(\mu \in I_n \text{ for all } 2 \leq n \leq \ell) = 1-\alpha.$$

It is of practical importance to know what this probability would be in the range, $\ell_1 \leq \ell \leq \ell_2$, with ℓ_1 and ℓ_2 finite integers. Indeed, Monte Carlo experiments showed that for $\ell_1 = 10$ and $\ell_2 = 80$ the real coverage probability seems to be considerably less than $1-\alpha$. Intuitively this probability would increase when the starting point of the sequence $\{I_n\}$ is increased to some $m > 2$. The evaluation of the coverage probability of the sequence for $m \leq n \leq \ell$ is analytically tedious (cf. the approximations used by Siegmund (1977)). For this reason Monte Carlo methods were used to obtain estimates of the probabilities. The following is a description of the method used: Generate independent $N(0;1)$ -random numbers by using the method of Chen (1971). Let $X_1, X_2, \dots, X_\ell$ be the generated sample of size ℓ . By repeating K Monte Carlo trials, the proportion of times

(p, say) that $u_n \leq n z_\alpha^2 \ell / (z_\alpha^2 \ell - n(1-n))$ for all n such that $m \leq n \leq \ell$, gives an estimate of the coverage probability of $\{I_n\}$, $m \leq n \leq \ell$ for given m , ℓ and α . Here

$$u_n = (\sum_{i=1}^n X_i)^2 / \sum_{i=1}^n X_i^2.$$

The estimated standard error of Monte Carlo estimation can be reduced by using the following technique:

Let t be the proportion of trials for which $u_k \leq k z_\alpha^2 \ell / (z_\alpha^2 \ell - k(1-k))$ for a fixed k , $m \leq k \leq \ell$ and τ be the (calculable) probability of this event. Instead of using p as an estimate of the coverage probability, rather use

$$p' = p + \gamma(t - \tau),$$

(which has the same expected value as p) and choosing γ such that the variance of p' be a minimum. The choice $\gamma = -\text{Cov}(p, t) / \tau(1-\tau)$ results in the minimum variance, namely

$$\text{Var}(p') = p(1-p)(1-\rho^2)/K$$

where ρ is the correlation coefficient between p and t .

As estimate for γ , $c = (w - pt) / (t(1-t))$ can be used, where w is the proportion of trials within which both the above-mentioned events occur.

Take as estimate of ρ , $r = c \{t(1-t) / p(1-p)\}^{\frac{1}{2}}$.

The success of this technique thus depends on the proper choice of an index k such that ρ is as large as possible. It is apparent that the estimate p' can be extended to

$$p' = p + \sum_{j=1}^J \gamma_j (t_j - \tau_j),$$

with each t_j corresponding to some index k , $m \leq k \leq \ell$.

For $K = 1000$ and $k = \ell$, the results are given in Table I for the cases $\alpha = 0,05$ and $\alpha = 0,01$.

The estimated coverage probabilities with their estimated standard errors (in brackets) are given together with the m -values corresponding to each α and ℓ -value.

In practice for given ℓ and $\alpha = 0,05$ or $0,01$ the appropriate m can be chosen using table I. Since $\{I_n\}$ is a $(1-\alpha)$ -confidence sequence, it follows from (1) that $\{J_n\} = \{\cap_{k=m}^n I_k\}$ is a $(1-\alpha)$ -confidence sequence as well. For interval estimation this sequence is at any stage shorter than or equal in length to $\{I_n\}$, and should be preferable. The only difficulty which arises is that it might happen that $J_{n'} = \emptyset$, the empty set, for some n' .

TABLE I

ℓ	$\alpha = 0,05$		$\alpha = 0,01$	
	m	Probability	m	Probability
16	15	0,9513 (0,0028)	—	
20	16	0,9434 (0,0044)	19	0,9882 (0,0010)
30	17	0,9489 (0,0042)	22	0,9882 (0,0017)
40	20	0,9441 (0,0049)	26	0,9903 (0,0014)
50	23	0,9556 (0,0037)	31	0,9909 (0,0014)
60	26	0,9441 (0,0051)	36	0,9862 (0,0026)
80	30	0,9501 (0,0046)	38	0,9896 (0,0020)

However, the probability of this event is at most α . In such a case one can construct a new $(1-\alpha)$ -confidence sequence as follows:

Either I_n' lies completely above $J_{n'-1}$ or completely below it. In the first case one identifies the r for which I_r is the interval with minimum upper bound among $I_m, \dots, I_{n'-1}$.

Substituting this bound by $+\infty$, the new sequence $\{J_n'\}$ which is included in the new I_r is seen to be a $(1-\alpha)$ -confidence sequence. Substitution of $-\infty$ for the maximum lower bound of I_s (say) in the other case will maintain the same result. This process can be repeated if necessary whenever an empty intersection appears.

3. A two-sided sequential t-test based on the confidence sequence for μ

Several sequential tests exist to test $H_0 : \mu = \mu_0$ against a two-sided alternative. Armitage (1975) describes in some detail an open sequential test due to G.A. Barnard, but unfortunately extended tables of the rejection and acceptance boundaries of this test are not widely available, which takes it out of reach for the practitioner. However, for the cases $\alpha = 0,05$ and $0,01$ and with power $1-\beta = 0,95$ under alternatives $\mu/\sigma = 0,2(0,1)1,0; 1,2$ and $1,4$ Armitage (1975) gives these boundaries. Using the same rejection boundaries as in the open plans, Schneiderman and Armitage (1962) suggested what they called restricted sequential plans as well as wedge-sequential plans. A condensed description of these tests may be found in Yuen (1977) and Armitage (1975).

Another approach for testing H_0 is based on the confidence sequence $\{I_n\}$. This test (two-sided) can be described as follows: First consider m observations, and then take sequentially observations until stage N , where $N = \min(\ell, N')$ and

$$N' = \min \{n \geq m: \mu_0 \notin I_n\}.$$

If $N' \leq \ell$, accept $H_1: \mu \neq \mu_0$, and if $N' > \ell$ accept H_0 .

The probability of type I error is easily seen to be at most α , since $P_\mu(\text{accept } H_1)$ for $\mu = \mu_0$ is equal to $P_{\mu_0}(\mu_0 \notin I_n \text{ for some } m \leq n \leq \ell) \leq \alpha$.

Robbins (1970) used the same approach for cases where $\ell = \infty$, which resulted in tests with power one. For $\ell < \infty$ nothing can be said in general about the power of such tests. For the test under consideration, Samuel-Cahn (1974b) suggested that the power (or the operating characteristic (OC)) of the proposed test can be taken approximately the same as that of the test where σ is known. Tables I and V of Samuel-Cahn (1974a)

with μ replaced by $\delta = (\mu - \mu_0)\ell^{1/2}/\sigma$ can be used not only to find the OC-function for different α -values, but also the average sample number (ASN) $E(N)$. We shall pay attention to the accuracy of this approximation in the last section.

Since this test has three parameters, namely the constant z_α , and the integers m and ℓ , it should be possible to select a value of ℓ for given α (and thus z_α and m fixed) to give a prescribed OC-value β under the alternative $\mu = \mu_1$.

From Samuel-Cahn (1974a) equation (1.9) we have for $\delta_1 = (\mu_1 - \mu_0)\ell^{1/2}/\sigma$ that

$$(4) \quad 1 - \beta = P_{\delta=\delta_1}((\bar{X}_n - \mu_0)/s_n \geq z_\alpha \ell^{1/2}/n \text{ for some } m \leq n \leq \ell)$$

$$\doteq 1 - \Phi(z_\alpha - \delta_1) + \exp(2\delta_1 z_\alpha) \Phi(-z_\alpha - \delta_1),$$

$$> 1 - \Phi(z_\alpha - \delta_1),$$

the approximate power of a two-sided non-sequential t test of size α when the sample size is large.

TABLE II

Values of $\delta_1 = (\mu_1 - \mu_0)\ell^{1/2}/\sigma$

$\alpha \backslash \beta$	0,10	0,05	0,02	0,01
0,10	3,24	3,52	3,86	4,09
0,05	3,60	3,88	4,22	4,45
0,01	4,29	4,57	4,91	5,14

To be on the safe side, our conjecture for the value of ℓ is approximately (using (4)):

$$\ell = \{(z_\alpha - \Phi^{-1}(\beta))\sigma/(\mu_1 - \mu_0)\}^2$$

Using this, Table II gives the values for δ_1 we propose to use for different α and β values.

Simulations of the OC-function in two cases (in the following section) support this conjecture about ℓ .

4. Properties of the sequential test

In the previous section we gave approximations which can be used to obtain the OC-function, ASN and maximum sample size of the proposed test. In order to judge the accuracy of the approximations, Monte Carlo trials (500 for each δ -value) were conducted. Apart from the OC-values for different δ -values, also the ASN ($E_\delta(N)$), the standard deviation ($\sigma_\delta(N)$), median ($Me_\delta(N)$), first- and third quartiles ($Q_\delta^{(1)}(N)$, $Q_\delta^{(3)}(N)$) of the sample size N were estimated to give an indication of its distribution.

We selected the following two-sided tests for $H_0 : \mu = 0$; with $\alpha = 0,05$:

- (a) Under the alternative $|\mu_1|/\sigma = 1$, with type II error probability $\beta = 0,05$. Using tables I and II, the appropriate m and ℓ values were chosen as 15 and $16^\dagger$ (Table III (a)).
- (b) Under the alternative $|\mu_1|/\sigma = 0,5$ with $\beta = 0,05$ and $m = 26$, $\ell = 60$ (Table III (b)).

Taking into account the standard errors in the second column, it is clear that the OC-values of the sequential tests agree well with those of the non-sequential tests (with sample sizes of 16 and 54) in both cases.

For an alternative as in (a) it can be advantageous to apply the sequential test instead of the standard non-sequential one, since saving of one observation is possible without loss in power.

For the alternative considered in (b), the non-sequential test will be better to use when $|\mu|/\sigma$ is near zero, but for $|\mu|/\sigma > 0,25$ it can be advantageous to use the sequential test, especially under far alternatives, where a saving of up to 52 per cent of the observations needed for the usual t-test, is possible.

[†]The value of ℓ should rather be 15, but since the corresponding non-sequential t-test requires 16 observations, the choice $\ell = 16$ seems more natural.

TABLE III

$ \mu /\sigma$	OC		$E_{\delta}(N)$ $\sigma_{\delta}(N)$ $Q_{\delta}^{(1)}(N)$ $Me_{\delta}(N)$ $Q_{\delta}^{(3)}(N)$				
	Sequential	Non-Sequential					
(a) 0,00	0,9513 (0,0028)*	0,950	15,96	0,20	16	16	16
0,25	0,8272 (0,0081)*	0,845	15,87	0,33	16	16	16
0,50	0,5062 (0,0110)*	0,535	15,62	0,49	15	16	16
0,75	0,1751 (0,0096)*	0,200	15,28	0,45	15	15	16
1,00	0,0344 (0,0056)*	0,038	15,06	0,25	15	15	15
1,25	0,0043 (0,0022)*	0,003	15,02	0,13	15	15	15
(b) 0,000	0,9441(0,0051)*	0,950	59,22	4,22	60	60	60
0,125	0,816 (0,017)	0,853	57,46	6,72	60	60	60
0,250	0,552 (0,022)	0,562	52,58	10,70	45	60	60
0,375	0,200 (0,018)	0,228	44,24	11,88	34	44	57
0,500	0,0492 (0,0068)*	0,050	35,55	9,70	27	33	42
0,625	0,006 (0,003)	0,005	30,85	6,99	26	27	34

*K = 1000, using the technique in Section 2

Next, we want to compare the proposed sequential test with existing sequential tests. The first test to compare with will be the RST-sequential plan which is based on the confidence sequence (2). From Table 6.4 of Armitage (1975) it follows that the appropriate plans to compare with the proposed test in cases (a) and (b) above, would be those having $\ell = 21$ and 78; and $\alpha' = 0,007$ and 0,0041. As nothing theoretically is known about the OC-function or the distribution of the sample size (denoted by N_R) of this test, we carried out simulations which are reported in Table IV.

Inspection of Table IV leads us to the following conclusion:

Although the OC-function of the RST-plan seems to agree closely with that of the proposed test in table III, considerably more observations are needed (up to 26 per cent more), except under far alternatives. Note also the large spread of the distribution of N_R . Another test to compare with the proposed test is the restricted sequential t-test of Schneiderman and Armitage (1962), since as the RST-plan, it has a truncation-point where H_0 is accepted as soon as the sample size exceeds that point.

Comparing Table III with tables 4 and 5 of Schneiderman and Armitage it is clear that the proposed test has greater power all the way and smaller ASN-values for alternatives near H_0 especially in case (a) where on the average the restricted test needs up to 29 per cent more observations than the proposed test. On the other hand, under far al-

ternatives the restricted test does much better than the proposed one, but with larger standard deviations for the sample sizes.

TABLE IV

(a) RST-plan, $\alpha = 0,05; \beta = 0,05, \mu_1 /\sigma = 1,0$						
$ \mu /\sigma$	OC	$E_\delta(N_R)$	$\sigma_\delta(N_R)$	$Q_\delta^{(1)}(N_R)$	$Me_\delta(N_R)$	$Q_\delta^{(3)}(N_R)$
0,00	0,942 (0,010)	20,15	3,59	21	21	21
0,25	0,874 (0,015)	19,46	4,60	21	21	21
0,50	0,580 (0,022)	17,18	5,93	14	21	21
0,75	0,286 (0,020)	13,92	6,47	8	14	21
1,00	0,034 (0,008)	9,86	5,11	6	9	13
1,25	0,004 (0,003)	8,05	4,05	5	7	10

(b) RST-plan, $\alpha = 0,05; \beta = 0,05, \mu_1 /\sigma = 0,5$						
0,000	0,946 (0,010)	74,68	14,37	78	78	78
0,125	0,850 (0,016)	71,02	19,24	78	78	78
0,250	0,590 (0,022)	62,95	22,56	50	78	78
0,375	0,214 (0,018)	46,71	25,19	25	46	75
0,500	0,054 (0,010)	34,70	21,57	17	31	49
0,625	0,006 (0,004)	23,71	14,50	13	21	32

TABLE V

Contaminated normal distribution,
 $\alpha = 0,05; \beta = 0,05; |\mu_1|/\sigma = 1,0$

$ \mu /\sigma$	Proposed test	Restricted test
0,0	0,956(0,007)	0,972
0,5	0,688(0,015)	0,750
1,0	0,190(0,012)	0,212
1,5	0,040(0,006)	0,025

Following Yuen (1977), the robustness of the proposed test can be investigated. To consider the influence of a small departure from normality, Monte Carlo trials were based on observations generated from a contaminated normal distribution. The same case was considered as that studied by Yuen, namely a 10 per cent contamination with a $N(\mu;9)$ -distribution, also for case (a) above (see Table V)

Comparing Tables V and III(a) it is clear that although the size of the test remains about the same, the OC-function is generally higher for $|\mu| > 0$. Compared with the restricted test under the same departure from normality (see Table V) the proposed test has a slightly lower OC-function.

To give a better picture of the influence which this departure has on the ASN-values of the proposed test, results of Monte Carlo experiments are displayed in table VI for case (b) above.

TABLE VI

Contaminated normal distribution,

$$\alpha = 0,05; \beta = 0,05; |\mu|/\sigma = 0,5$$

$ \mu /\sigma$	OC	$E_{\delta}(N)$	$\sigma_{\delta}(N)$	$Q_{\delta}^{(1)}(N)$	$Me_{\delta}(N)$	$Q_{\delta}^{(3)}(N)$
0,00	0,958(0,009)	59,61	2,23	60	60	60
0,25	0,688(0,021)	55,27	9,12	56	60	60
0,50	0,172(0,017)	42,14	12,21	31	41	53
0,75	0,026(0,007)	32,23	8,48	26	29	36

In addition, the effect of departure from independence was investigated empirically by Monte Carlo methods in the same way as was done by Yuen. Observations were generated sequentially by

$$X_{i+1} = \rho X_i + e_{i+1}, \quad i = 1, 2, \dots;$$

where X_1 is distributed $N(0;1)$ independently from the e_i 's, which are independent $N(0;1-\rho^2)$. Choosing $\rho = 0,1$, the proposed test for cases (a) and (b) has sizes 0,07(0,008) and 0,0816(0,009) respectively and therefore cannot be recommended in a situation of even slightly dependent observations.

5. Application of the proposed test

The fact that the proposed test seems to tolerate small departures from normality, and the saving of observations under alternatives not too near to H_0 , make this test a strong competitor of the usual t-test. We also saw that the restricted sequential t-test requires considerably more observations to draw inference for δ -values near H_0 . From the tables in Schneiderman and Armitage mentioned above as well as those of Yuen, it seems that the wedge-sequential t-test has very appealing properties as far as its robustness and the sample sizes under all alternatives are concerned. The question arises whether this test would not be better for application in practice. However in our opinion there are two reasons why we would rather recommend the proposed test:

- (a) The rejection boundaries for different values of α and β of the wedge-sequential t-test are tabulated but not widely available. On the other hand, the proposed test needs no tables to apply, except for the choices of m and ℓ , the initial and maximum sample sizes.
- (b) Since the proposed test is based on a confidence sequence, a $(1-\alpha)$ -confidence interval can be obtained for μ at any stage. This is usually important when the test is terminated (see the next section). No such confidence interval can be based on the wedge-sequential t-test, not even an ordinary $t(1-\alpha)$ -confidence interval when the test is terminated (the coverage probability of such an interval is less than $1-\alpha$ since it also depends on the stochastic sample size).

To conclude, the accuracy of the approximations to the OC and ASN-functions of the proposed test when using the tables of Samuel-Cahn (as was mentioned in Section 3) can be judged in the light of the two cases which were discussed in the previous section.

TABLE VII
Approximate OC- and ASN-values given by Samuel-Cahn;
 $\alpha = 0,05; \beta = 0,05$

(a) $ \mu_1 /\sigma = 1,0$			(b) $ \mu_1 /\sigma = 0,5$		
$ \mu /\sigma$	OC	ASN	$ \mu /\sigma$	OC	ASN
0,00	0,950	15,82	0,000	0,950	59,34
0,25	0,838	15,39	0,125	0,845	57,83
0,50	0,509	15,00**	0,250	0,532	52,33
0,75	0,178*	15,00**	0,375	0,272*	47,76*
1,00	0,029*	15,00**	0,500	0,042*	34,85*
1,25	0,002*	15,00**	0,625	0,005*	28,12*

The values were obtained from Table V, Samuel-Cahn (1974a) by interpolation between δ -values. For larger δ -values Table I was used (indicated by “*”) where interpolation also between α -values was necessary. Where ASN-values from the tables were less than m , the value of m was used instead (indicated by “**”).

Comparing Table VII with Table III, we see that the approximations give a good indication of the “true” OC- and ASN-functions. As can be expected, the approximation of the ASN-values under far alternatives is lower than the “true” sample sizes, since the proposed test takes an initial sample which is not the case for the test for which the tables were computed.

In practice, the implementation of Samuel-Cahn’s tables therefore seems to be the way to calculate OC- and ASN-values for different α and β -values (by using also tables I and II above).

6. A sequential confidence interval

From the definition of a $(1-\alpha)$ -confidence sequence, it is possible to construct a sequence of intervals such that any one of these intervals has a coverage probability of $1-\alpha$. This can be useful in situations where the client requires a specified accuracy for the interval estimate of μ . Without knowing σ it is not possible for the statistician to calculate the required sample size beforehand. Using a $(1-\alpha)$ -confidence sequence, the client can inspect the interval at every stage after the initial m observations were taken, and he can subjectively decide when enough observations were made (he can even alter his initial criterion for accuracy when the intervals show that σ is much larger than has been expected).

In the previous section the possibility of computing a $(1-\alpha)$ -confidence interval for μ after the sequential test for $H_0: \mu = 0$ has terminated was pointed out (especially when H_0 was rejected). It is only necessary to evaluate the sequence $\{J_n\}$ (as was seen in Section 2) for $n \geq m$, checking whether it includes zero rejecting H_0 if not and using J_N as the $(1-\alpha)$ -confidence interval for μ .

For the cases (a) and (b) described in Section 4, we calculated the average upper and lower bounds for J_N . The results are displayed in Table VIII.

The expected lower and upper bounds of the non-sequential 0.95-confidence intervals based on 16 and 54 observations respectively, are: $(-0.5328; 0.5328)$ and $(-0.2582; 0.2582)$, when $\mu = 0$.

The sequential intervals under $\mu = 0$ are 3.3 per cent and 7 per cent wider than those of the corresponding non-sequential intervals, which is quite a small price to be paid for greater flexibility. On the other hand, under the alternatives the sequential intervals are 12 per cent and 98 per cent wider, which could have been expected since they are based on fewer observations than the non-sequential intervals.

The comparison of J_N with the non-sequential interval was made only to put it into perspective. No non-sequential $(1-\alpha)$ -confidence interval can be combined with a sequential test and still have the prescribed coverage probability.

TABLE VIII

0,95-confidence interval bounds for μ , when $\sigma = 1$, after the test
 $H_0 : \mu = 0$ has terminated;

(a) $ \mu_1 = 1,0$			(b) $ \mu_1 = 0,5$		
$ \mu $	Lower	Upper	$ \mu $	Lower	Upper
0,00	-0,5335	0,5668	0,000	-0,2836	0,2693
0,25	-0,2765	0,8351	0,125	-0,1452	0,4338
0,50	-0,0404	1,0926	0,250	-0,0548	0,6107
0,75	0,2217	1,3913	0,375	0,0047	0,8285
1,00	0,4784	1,6675	0,500	0,0474	1,0681
1,25	0,7220	1,9150	0,625	0,0911	1,2482

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